Supplementary Material for “A Homogeneous Compensation Mechanism for Indirect Evolutionary Rescue in a Roy-Style Predator–Prey Model”

Mathbio research notes

May 31, 2026

S1. Numerical classification protocols

The main manuscript reports consolidated outputs from the numbered analysis scripts rather than new simulations. Classification used the same vocabulary across ODE and PDE outputs:

- **Persistent**: predator density remains positive and late-window change/residual diagnostics indicate a steady or effectively settled state.
- **Extinct**: predator density approaches the extinction threshold and late-window diagnostics do not indicate recovery.
- **Transient**: the trajectory does not meet persistent or extinct steady criteria within the tested horizon, often because late-window change or residual diagnostics remain non-negligible.

The tail-window logic was used to avoid treating long transients as equilibria. For PDE tests, classification of spatial means was paired with spatial coefficient-of-variation diagnostics for n, w, and q. A basin label was compared with a matched homogeneous control when heterogeneous perturbation effects were evaluated. The classification logic is diagnostic, not a proof of global basin structure.

S2. Mathematical appendix: ODE Jacobian and Routh–Hurwitz margins

Write

$$F_n = nR, \quad R = r(q)z - \xi - a(q)w,$$

$$F_w = wP, \quad P = b(q)nz - (m+s) - \mu w,$$

$$F_q = \nu q(1-q)G, \quad G = \Delta r z - \Delta a w,$$

with $z = \kappa^{-1} - n - w$, $\Delta r = r_v - r_u$, $\Delta a = a_v - a_u$, and $\Delta b = b_v - b_u$. Since $z_n = z_w = -1$,

$$R_n = -r(q), \qquad R_w = -r(q) - a(q), \qquad R_q = G,$$

$$P_n = b(q)(z-n), \qquad P_w = -b(q)n - \mu, \qquad P_q = \Delta b\, nz,$$

$$G_n = -\Delta r, \qquad G_w = -\Delta r - \Delta a, \qquad G_q = 0.$$

The full reaction Jacobian in variables (n, w, q) is therefore

$$J = \begin{pmatrix} R - nr(q) & n[-r(q) - a(q)] & nG \\ w\, b(q)(z-n) & P + w[-b(q)n - \mu] & w\,\Delta b\, nz \\ -\nu q(1-q)\Delta r & -\nu q(1-q)(\Delta r + \Delta a) & \nu(1-2q)G \end{pmatrix}.$$

At an interior compensation-branch equilibrium, $R = P = G = 0$, giving the simplified branch Jacobian

$$J_* = \begin{pmatrix} -n^* r(q^*) & -n^*[r(q^*) + a(q^*)] & 0 \\ w^* b(q^*)(z^* - n^*) & -w^*[b(q^*)n^* + \mu] & w^* \Delta b\, n^* z^* \\ -\nu q^*(1-q^*)\Delta r & -\nu q^*(1-q^*)(\Delta r + \Delta a) & 0 \end{pmatrix}.$$

For $\det(\lambda I - J) = \lambda^3 + A_1\lambda^2 + A_2\lambda + A_3$, the coefficients were computed as

$$A_1 = -\mathrm{tr}(J), \qquad A_2 = \frac{1}{2}\left[(\mathrm{tr}J)^2 - \mathrm{tr}(J^2)\right], \qquad A_3 = -\det(J).$$

Table 1 reports the Routh–Hurwitz terms and the minimum margin at the target stresses.

Table 1: Routh–Hurwitz margins for the linear compensation branch at target stresses. Values are copied to `results/roy_ode_compensation_routh_hurwitz_margins.csv`.

s	A_1	A_2	A_3	$A_1A_2 - A_3$	min margin
0	3.9468	0.33673	0.003540	1.3255	0.003540
0.069448242	4.3241	0.27528	0.004006	1.1863	0.004006
0.11765625	4.5861	0.21486	0.003566	0.98178	0.003566
0.1584375	4.8077	0.15238	0.002706	0.72989	0.002706
0.16486816	4.8426	0.14158	0.002529	0.68308	0.002529
0.175	4.8976	0.12403	0.002228	0.60525	0.002228

S3. Additional ODE basin maps

The main text focuses on the derived compensation branch and its stability. Additional ODE diagnostics support the basin interpretation and the homogeneous reaction-level mechanism. Figure 1 shows representative ODE time series and selection/growth diagnostics. Figure 2 shows the ODE q_0–w_0 basin map used to relate initial defense and predator density to persistent, extinct, and transient outcomes. Figure 3 shows selection-growth phase diagnostics and numerical equilibrium stability.

S4. PDE non-homogeneous perturbation diagnostics

The main manuscript reports the conclusion from spatial-mode stability and long-horizon follow-up. The supporting PDE diagnostics are shown here. Figure 4 summarizes additional spatial-stability views. Figure 5 shows the continuous λ-scan over the mode-equivalent range $i, j = 0, \ldots, 64$. Figure 6 shows representative non-homogeneous initial and final fields. Figure 7 shows mean dynamics and spatial coefficient-of-variation diagnostics for the targeted non-homogeneous tests. Figure 8 shows the long-horizon follow-up diagnostics for the finite-horizon basin-changing cases.

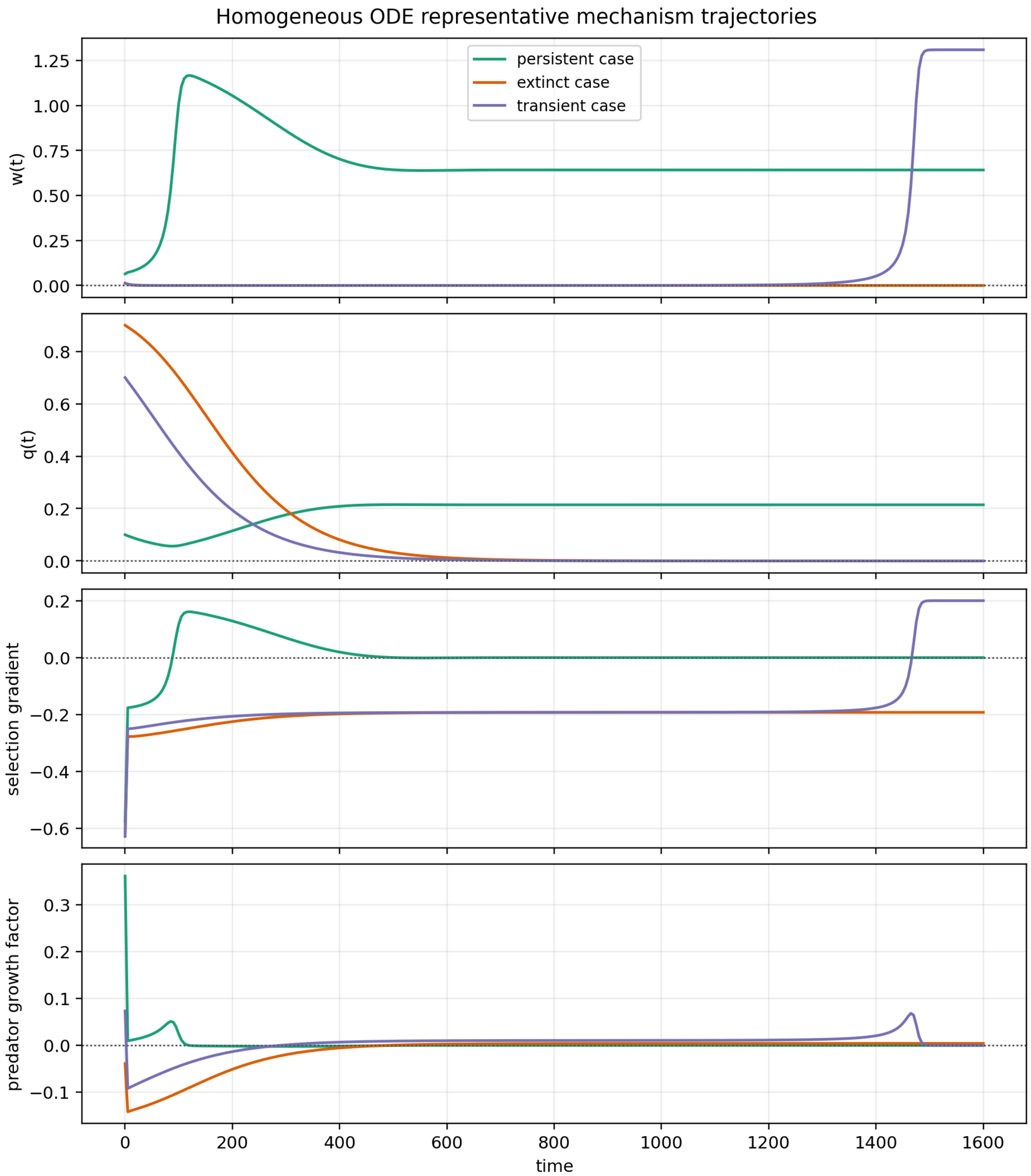

Figure 1: Representative ODE trajectories for predator density, defense frequency, selection-gradient diagnostics, and predator-growth diagnostics. These panels support the homogeneous mechanism interpretation but do not replace the analytic branch derivation.

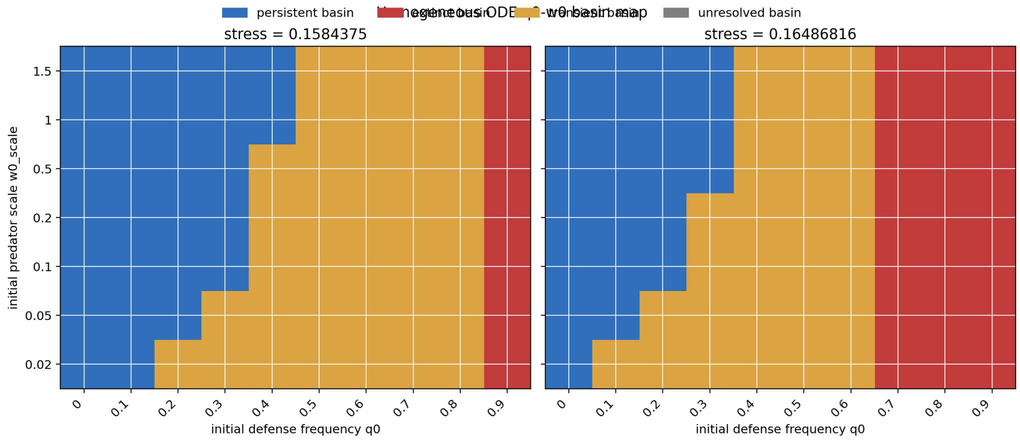

Figure 2: ODE basin map on the targeted q_0–w_0 grid. This figure supports basin dependence in the homogeneous reaction system and is not an exhaustive basin-boundary refinement.

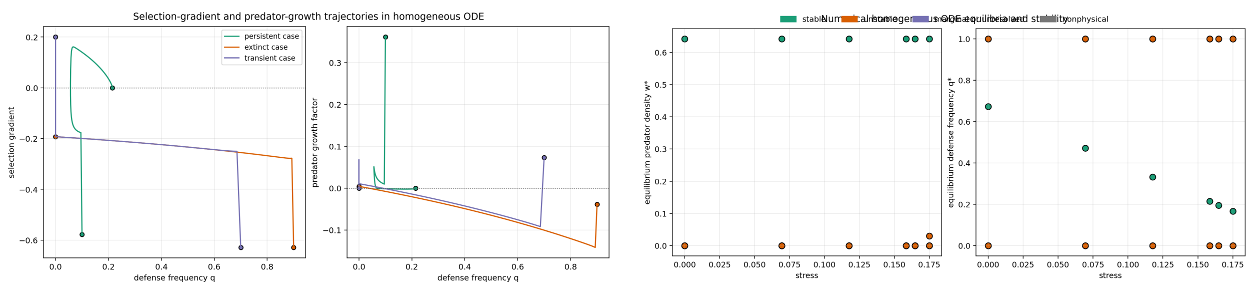

Figure 3: Additional homogeneous mechanism diagnostics. Left: selection-gradient and predator-growth relationships for representative cases. Right: numerical equilibria and stability labels across tested stresses.

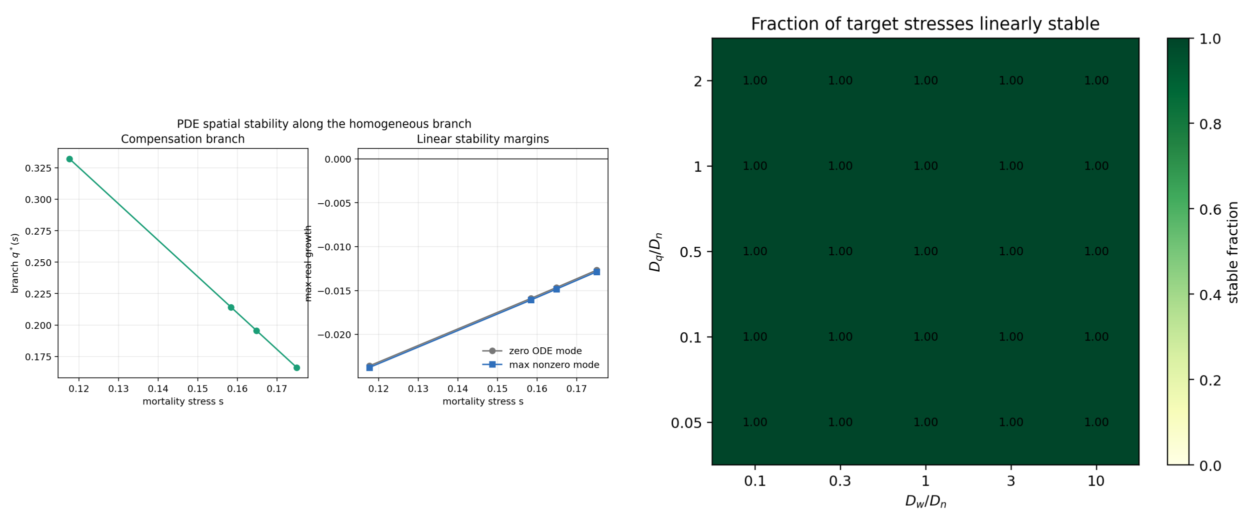

Figure 4: Additional linear PDE spatial-stability diagnostics. These figures support stability for the tested branch states and diffusion-ratio grid, not a theorem over all diffusion coefficients.

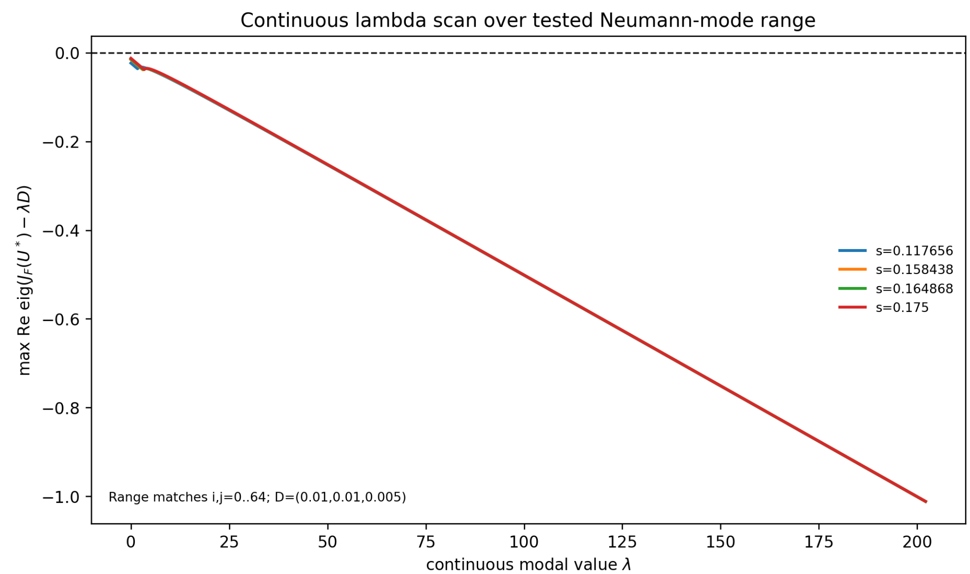

Figure 5: Continuous λ-scan for $J_F(U^*) - \lambda D$ over the mode-equivalent range corresponding to $i, j = 0, \ldots, 64$. This scan strengthens the tested spatial-mode result over that range, but it is still a tested-range diagnostic rather than a proof for all possible modes or diffusion coefficients.

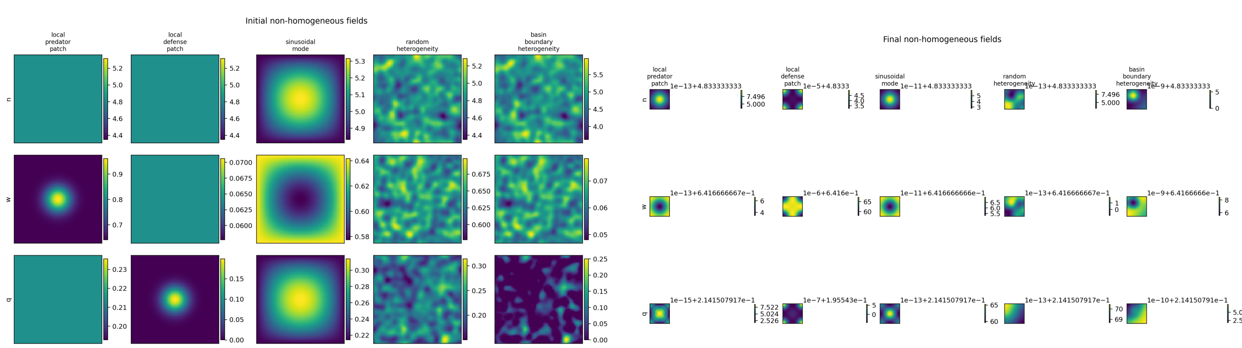

Figure 6: Representative non-homogeneous initial and final PDE fields. These diagnostics show the tested perturbation classes and final-field behavior for targeted cases.

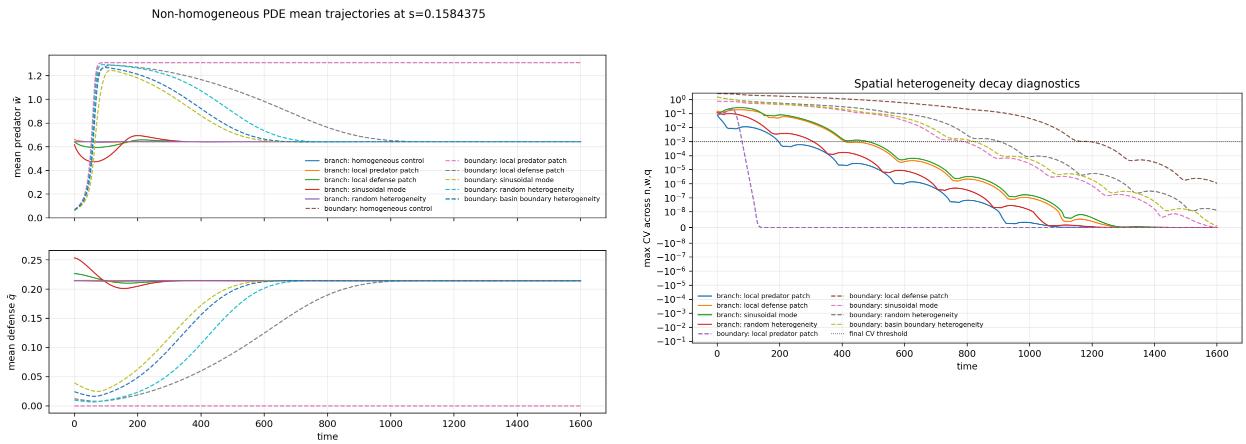

Figure 7: Mean dynamics and spatial coefficient-of-variation diagnostics for targeted finite-amplitude non-homogeneous PDE tests.

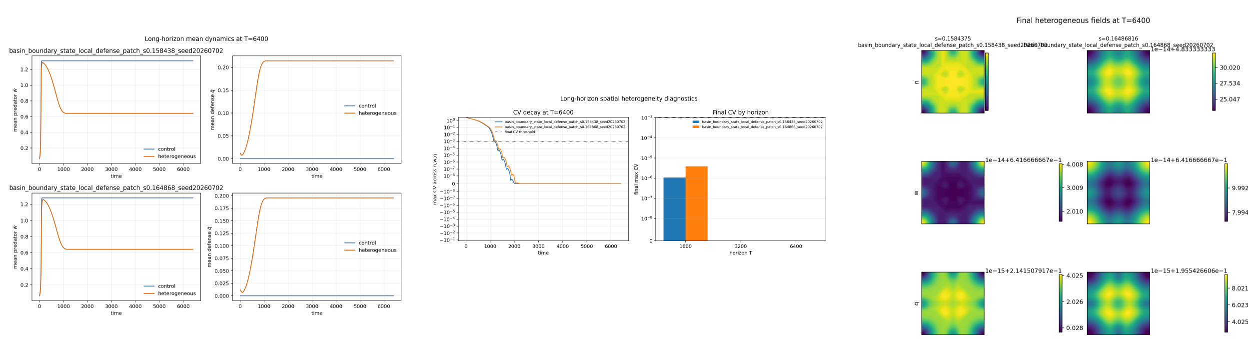

Figure 8: Long-horizon follow-up diagnostics for finite-horizon basin-changing non-homogeneous cases. These diagnostics support resolution to matched homogeneous controls without persistent spatial patterning for the followed cases.

S5. Nonlinear trade-off shape-grid details

The nonlinear extension uses endpoint-preserving shape functions $f_\gamma(q) = q^\gamma$ with $\gamma \in \{0.5, 1, 2\}$. The purpose is to test whether the compensation mechanism survives selected concave, convex, and mixed trade-off shapes, not to scan all possible nonlinear functions. Figure 9 shows selected nonlinear branch curves and shape-grid summaries. Figure 10 shows selected nonlinear PDE spatial-stability and non-homogeneous perturbation diagnostics.

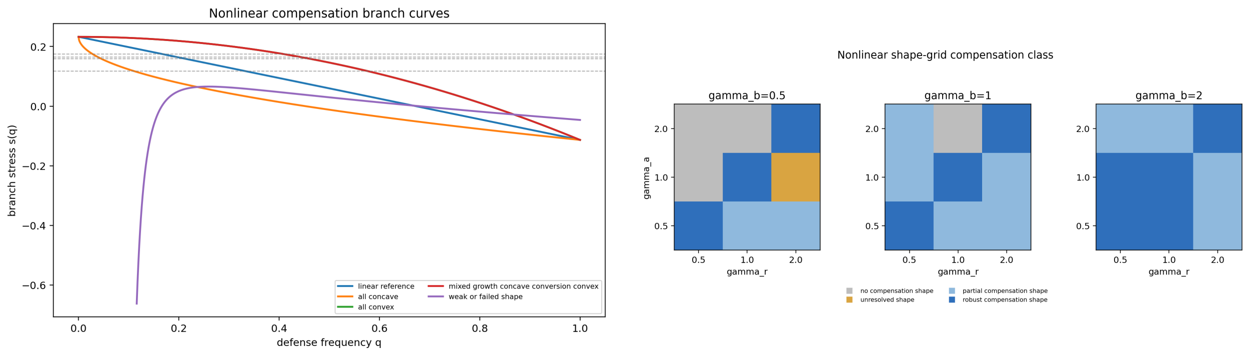

Figure 9: Nonlinear branch curves and shape-grid summary. These figures support selected nonlinear compensation branches within the controlled grid.

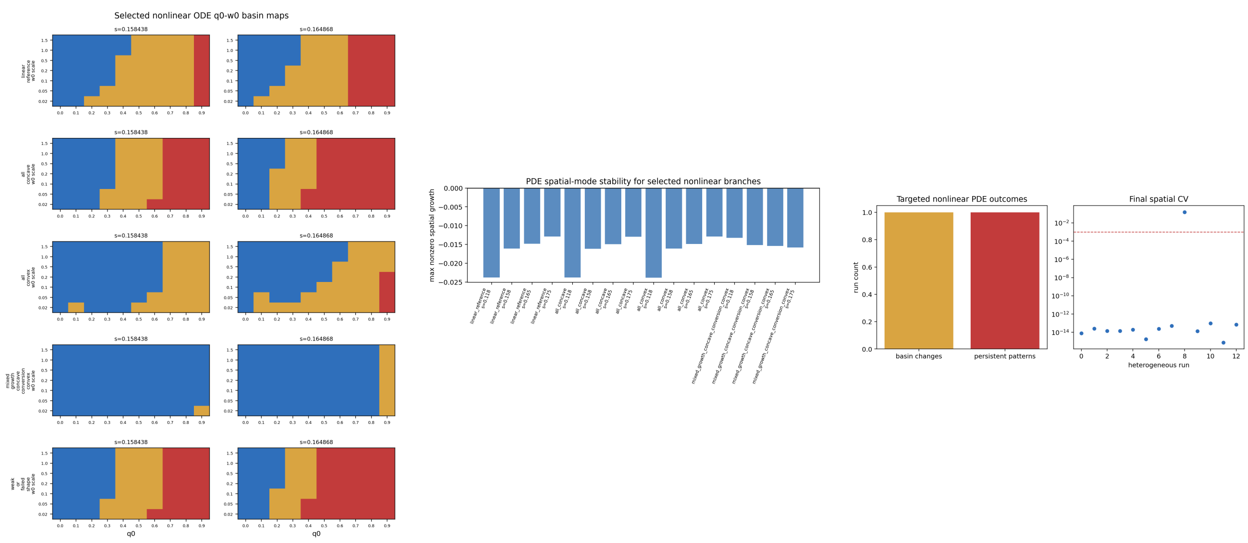

Figure 10: Selected nonlinear ODE and PDE diagnostics. These panels are supporting evidence for controlled nonlinear extension and do not establish global nonlinear-trade-off robustness.

S6. Additional figure list and CSV source mapping

Table 2 maps the main analysis scripts to the CSV outputs used by the manuscript package. Table 3 maps the main claims to source CSVs and representative figures. The status column records whether each item is a main result or supporting diagnostic evidence.

Table 2: Script-to-CSV mapping for reproducibility.

Script	CSV outputs used here
`experiments/23_roy_ode_compensation_conditions.py`	`roy_ode_compensation_conditions_summary.csv`; `roy_ode_compensation_condition_grid.csv`
`experiments/24_roy_ode_compensation_routh_hurwitz.py`	`roy_ode_compensation_routh_hurwitz_current.csv`; `roy_ode_compensation_routh_hurwitz_summary.csv`
`experiments/25_roy_pde_spatial_stability_and_pde_compensation_tests.py`	`roy_pde_compensation_spatial_modes_current.csv`; `roy_pde_spatial_nonlinear_mechanism_decision.csv`
`experiments/26_roy_pde_nonhomogeneous_long_horizon_followup.py`	`roy_pde_nonhomogeneous_long_horizon_decision.csv`; long-horizon summary and time-series CSVs
`experiments/27_roy_nonlinear_tradeoff_compensation.py`	`roy_nonlinear_tradeoff_compensation_decision.csv`; `roy_nonlinear_tradeoff_shape_summary.csv`; `roy_nonlinear_tradeoff_branch_recovery_linear.csv`
`experiments/28_roy_counterfactual_and_spatial_evidence.py`	`roy_ode_no_evolution_counterfactual_summary.csv`; stress-sweep and time-series CSVs; `roy_pde_compensation_lambda_scan_summary.csv`; `roy_ode_compensation_routh_hurwitz_margins.csv`

Table 3: Claim-to-source mapping for the manuscript package.

Claim	Source CSV	Representative figure	Status
Frozen-q counterfactual verifies indirect rescue over a finite stress window	`roy_ode_no_evolution_counterfactual_summary.csv`; stress-sweep CSV	Main no-evolution figure	Main claim
Linear compensation branch conditions are satisfied in the current parameterization	`roy_ode_compensation_conditions_summary.csv`	Fig. 3; main branch figure	Main claim
Routh–Hurwitz stability holds at target stresses	`roy_ode_compensation_routh_hurwitz_summary.csv`	Main Routh–Hurwitz figure	Main claim
Spatial modes are stable for the current PDE coefficients over the tested range	`roy_pde_spatial_nonlinear_mechanism_decision.csv`; `roy_pde_compensation_lambda_scan_summary.csv`	Main spatial-stability figure; Fig. 4; Fig. 5	Main claim
Long-horizon non-homogeneous basin changes resolve to homogeneous controls	`roy_pde_nonhomogeneous_long_horizon_decision.csv`	Main long-horizon decision panel; Fig. 8	Main claim
Linear branch is recovered by nonlinear branch formulation	`roy_nonlinear_tradeoff_branch_recovery_linear.csv`	Fig. 9	Supporting
Nonlinear shape grid includes robust, partial, no-compensation, and unresolved cases	`roy_nonlinear_tradeoff_shape_summary.csv`; `roy_nonlinear_tradeoff_compensation_decision.csv`	Main nonlinear decision figure; Fig. 9	Main claim

The main manuscript contains the primary compensation branch and stress-interval figure. This boxed note avoids duplicating that main-text figure in the supplement while keeping the claim-to-source map readable.

Figure 11: Pointer to the main compensation branch figure.